Victor Ivanov

Senior Full-Stack Engineer
Virginia · U.S. Citizen · Active DoD Secret clearance · victor.n.ivanov@gmail.com
victorivanov.engineer · github.com/LLRHook · LinkedIn

Secret-cleared, backend-leaning full-stack engineer with 4+ years shipping high-stakes certification SaaS. Owned a React/Spring Boot scheduling workflow end to end, carried a billing hotfix through live production validation, and built Next.js analytics and quiz experiences with explicit async, accessibility, and security controls. Open to on-site and hybrid roles and customer travel.

Experience

Senior Backend Engineer

May 2024 – present
Paradigm Testing · Remote

- Lead architecture, priorities, and production readiness for three engineers across four certification SaaS products.
- Owned a React/Spring Boot written-exam scheduling workflow end to end: create/edit wizards, dirty state, an ADA accommodation picker, structured 409 conflict handling, tenant-scoped CRUD, and 10 dedicated test files/classes.
- Owned a billing hotfix from root cause through protected production deployment; live reconciliation corrected an undercharge and restored accurate invoicing across a large document batch.
- Diagnosed a three-month multithreaded AWS file-lock failure; cut errors from 80% to near-zero for 10 organizations.
- Led a monolith-to-microservices rewrite into five Spring Boot 3 services for oral/written exams, plagiarism detection, AI proctoring, and results review; the MVP serves production traffic.
- Architected secure real-time exam video delivery with WebSockets/STOMP, direct-to-S3 uploads, JWT role controls, audit logs, and Valkey-backed refresh tokens.
- Partnered with a proctoring vendor and InfoSec to integrate proctoring across oral and written exam flows; authored cross-role acceptance steps and completed validation.
- Built merge-gated CI/CD and 550+ automated tests from zero; containerized Playwright auth, admin CRUD, and impersonation flows in Bitbucket Pipelines, supporting six releases without covered-workflow regressions.
- Turned live V&V and demo findings into a fail-closed, 10-stage pilot gate spanning tenant onboarding, candidate disconnect/resume, scoring, and audit reconstruction; exposed an answer-key leak before go-live.

Software Developer

June 2022 – May 2024
Paradigm Testing · Maryland

- Built a high-stakes oral-exam platform serving 4,800+ candidates annually; led JPA adoption across 73 entities and eliminated peak-load waiting-room errors at 200+ concurrent sessions.

Full-Stack Engineering Practicum

July – August 2026
Revature

- Delivered 63 merged pull requests in 22 days, including a Next.js trainer analytics dashboard with server-driven filters/paging, an accessible table equivalent for SVG analytics, independent async states, retry UX, and 27 Vitest cases.
- Rebuilt the quiz flow as a server-rendered Next.js route backed by FastAPI sessions: one request per question, zero-request history navigation, ownership-checked resume, token redaction, and Playwright E2E through nginx.

Selected projects

Citybase

Electron, React, Node.js

- Desktop IDE mapping repositories into an isometric city and dispatching Claude/Codex through typed IPC; streamed, cancellable runs, persisted history, guarded Git operations, and 393 tests.

Fix YouTube

JavaScript, Manifest V3

- Zero-dependency extension for SPA-aware Shorts redirects, feed filters, focus timers, watch history, and settings import/export; cross-browser CI and 15 tests.

Education

Georgia Institute of Technology

M.S. Computer Science (online)
Expected December 2027

University of Maryland, Baltimore County

B.S. Computer Science

Technical skills

Frontend: TypeScript/JavaScript, React, Next.js 16, Vite, HTML/CSS, React Query, Playwright, Vitest
Backend: Java 21, Spring Boot 3, Spring Security, JPA/Hibernate, Python, FastAPI, Go, REST, WebSockets/STOMP
Data & cloud: PostgreSQL, MariaDB, Redis/Valkey, AWS (EC2, S3, RDS), Docker, Kubernetes/Helm
Delivery & security: Bitbucket Pipelines, GitHub Actions, CI/CD, OAuth2/JWT, JUnit/Mockito, Testcontainers, Pytest
Updated October 2026.